

The digital transformation of industrial systems continues to advance, making the exchange of consistent and meaningful information an essential component for modern manufacturing. With the increasing number of digital tools and the variety of data sources, the need for standardization and semantic interoperability has become critical for efficient manufacturing processes /1/.

A core element in this transformation is the Asset Administration Shell (AAS), which functions as the standardized digital representation of an asset. It provides a structured, interoperable source of information that allows different systems to access and interpret asset data consistently /2/. In this way, the AAS can be utilized for data exchange, lifecycle management, and the monitoring of industrial components across diverse environments /2/.

In many industries, the handling of heterogeneous data remains a significant challenge. Different systems, manufacturers, and technologies often use varying definitions, terminologies, and structures to represent the same concepts or properties. This inconsistency in data representation can lead to interoperability issues, as seamless communication between systems becomes difficult.

In the case of AAS, such inconsistencies emerge when different manufacturers define or name the properties of their asset submodels in different ways. This lack of a uniform approach directly impacts their interoperability, preventing effective data transfer between various industrial systems and hindering the potential for a fully integrated, smart manufacturing environment /3/.

The ECLASS semantic dictionary system, which aims to standardize terminology across industries, has been discussed as a potential solution to this issue. ECLASS provides a comprehensive set of standardized attributes and descriptors for industrial products. However, it is not yet widely used in naming properties within the AAS submodels, largely due to the challenges involved in manually aligning existing datasets with this semantic dictionary. The manual effort required to map diverse, manufacturer-specific terms to ECLASS concepts is labor-intensive and prone to errors, ultimately limiting its effectiveness in practice /3/.

This is where the application of Large Language Models (LLMs), a powerful subset of Natural Language Processing, offers a promising solution. LLMs, such as GPT-3, BERT, and other advanced models, are capable of understanding and generating human language with remarkable precision and contextual awareness /4/, /5/. These models are well-suited for interpreting complex terminology and semantic relationships within technical texts.

The application of LLMs extends beyond specific use cases and addresses the challenge of heterogeneous data across industries, bridging the gap between diverse data sources by recognizing equivalent meanings across various terminologies and mapping them to standardized properties, thereby ensuring seamless interoperability (see Figure 1).

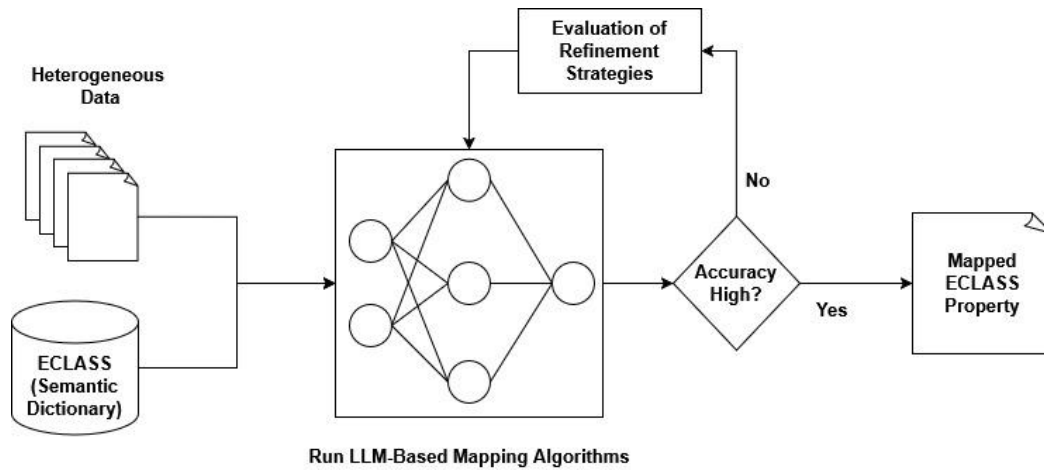


Figure 1: LLM-Based Semantic Mapping of Industrial Properties to ECLASS

In the context of AAS, this capability can be utilized to automatically map the diverse property labels of AAS submodels to common standards, such as ECLASS, reducing manual effort and ensuring consistent data representation.

In this paper, we propose a novel concept that utilizes LLMs to map heterogeneous industrial data from different manufacturers to descriptors defined by systems like ECLASS. The approach aims to establish a semantic mapping framework that automatically and accurately reconciles inconsistent terminology, enabling seamless data exchange and interoperability between different industrial systems. By automating this process, our method not only reduces manual intervention but also enhances the efficiency and scalability of data-driven industrial applications. As a result, it can also be applied to the use case of ensuring interoperability between AAS properties across different industries, while still addressing the broader challenge of heterogeneous data transfer within the industry.

Our research lays the groundwork for a future where interoperable and data-driven manufacturing systems can transfer data more effectively, helping to optimize industrial processes and facilitate smarter, more automated production environments. This work represents an important step toward overcoming one of the key barriers in the digitalization of manufacturing and advancing the vision of Industry 4.0.

Self-Assessment:

The proposed paper presents a novel conceptual framework for enhancing semantic interoperability in Asset Administration Shells using Large Language Models (LLMs). While existing work, such as the dissertation by Both, has laid important groundwork by describing the structure and principles of the AAS and highlighting the core challenge of semantic interoperability, it primarily focuses on static ontologies and rule-based approaches /1/. However, there remains a gap in dynamically resolving heterogeneous terminology using data-driven methods. The paper builds upon foundational research on AAS from Both's dissertation and extends it with state-of-the-art Artificial Intelligence capabilities. The preferred form of presentation is a conference talk. The contribution aligns with the CfP topic "Digital Transformation".

- /1/ Plattform Industrie 4.0 (2023): Handlungsfelder 2030, URL: <https://www.plattform-i40.de>
- /2/ IDTA (2023): Detailspezifikation der Verwaltungsschale, Version 3.0, Industrial Digital Twin Association.
- /3/ Both, M. (2024): Semantische Interoperabilität digitaler Zwillinge als Basis für automatisierte Erkundungsmechanismen, Dissertation, Otto-von-Guericke-Universität Magdeburg.
- /4/ Brown, T., Mann, B. et al. (2020): Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer, Journal of Machine Learning Research, Vol. 21, pp. 1–67. DOI: <https://doi.org/10.48550/arXiv.1910.10683>
- /5/ Raffel, C., et al. (2020): Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer, Journal of Machine Learning Research, Vol. 21, pp. 1–67. DOI: <https://doi.org/10.48550/arXiv.1910.10683>